Reading into Writing – Climate Change

Why is the scientific consensus on climate change often misunderstood or ignored by the public? What can be done about this?

Extract One:

Adapted fom: Maertens, R., Anseel, F., & van der Linden, S. (2020). Combatting climate change misinformation: Evidence for longevity of inoculation and consensus messaging effects. Journal of Environmental Psychology, 70, 101455.

Climate change is one of the most pressing problems of our time, requiring large-scale collective action. If no public action is taken, the continued rise in global temperatures could bring fundamental harm to human society and ecosystems. Numerous detrimental effects are already occurring, from extreme temperatures and floods to failed crop harvests and climate refugees. Indeed, many ecological systems are currently being threatened with destruction and biodiversity is falling drastically.

Among climate scientists, a strong scientific consensus has been established on the fact that humans are causing global warming. Consensus research has replicated this finding in at least five high-quality studies over the past years, estimating the consensus among scientists from a minimum of 91% to a maximum of 100%. These studies further show a strong positive relationship between expertise in climate science and scientific consensus on human-caused climate change. However, most people are not climate science experts, and need to navigate facts through a cloud of (mis)information.

1.1. Consensus messaging

Research has shown that communicating descriptive norms, such as the fact that 97% of scientists agree that humans are causing climate change, can positively influence belief in climate change and support for action, bridging the ideological divide. Although research on the benefits of communicating scientific consensus is now well-established, research has also shown that when it is countered with an opposing (misinformation) message that contradicts the scientific consensus (e.g., a misleading petition claiming “there is no scientific consensus on climate change”), the positive effect of communicating the consensus is reduced or even completely neutralised.

1.2. Misinformation

Efforts to tackle the climate change problem have suffered from the influence of various forms of misinformation. Among the many climate change misinformation techniques used, the most prevalent technique is sowing doubt about the scientific consensus. Through false-balance media coverage of the topic (e.g., 50% / 50% instead of 97% / 3%) or by using fake expert accounts (e.g., a professor in an unrelated field proclaiming to be a climate expert), perceptions of scientific consensus can be distorted.

At the same time, researchers find that online misinformation can spread faster and deeper than factual information, making it harder for scientific facts to reach the entire population. Even when factual information corrects a myth, the initial belief (based on a falsehood) can still exert a continued influence. When the correction goes against the prevailing worldview of a polarised group, this correction may even backfire, though it should be noted that the literature on backfire effects is increasingly debated. In short, although the 'debunking' approach to misinformation is often ineffective, it is still the most prevalent method used to tackle fake news. In contrast, it has been suggested that a more effective way of tackling misinformation is one based on building up resistance *before* any damage is done, an approach called "inoculation" or "pre-bunking".

Over the past 50 years, inoculation theory has emerged "as the most consistent and reliable method for conferring resistance to persuasion". Initially designed as a vaccine against brainwashing by propaganda, inoculation theory is now arguably the most established theory on resistance against persuasion. The inoculation process consists of two related components, namely: a) forewarning (alerting people in advance to the possibility of misinformation) and b) refutation (providing arguments or evidence to counter misinformation). The underlying idea is that prior experience with persuasive arguments, combined with a warning of an upcoming threat, provides familiarity and alertness which can be used to disarm an attack. Meta-analyses have demonstrated that this method is generally effective, but there is still a lack of clarity about the long-term effectiveness of inoculation effects. In most studies, the delay between the inoculation and the persuasion attempt is only a matter of minutes or at most a few days.

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Extract Two:

Adapted from: Davis, C. J., & Lewandowsky, S. (2022). Thinking about climate change: look up and look around!. Thinking & Reasoning, 28(3), 321-326.

Is humanity capable of responding adaptively to catastrophic global threats such as climate change or (as in the film Don't Look Up; McKay, 2021) approaching comets? To what extent is our response to such threats compromised by our thoughts and beliefs? Adam McKay, the director of Don't Look Up, has described it as 'a disaster movie in which people don't necessarily believe that the disaster is coming. Acknowledging anthropogenic climate change implies the need for radical changes to individual behaviour as well as the way societies are structured, which may conflict with one's value system.

There is also a financial cost of climate change mitigation. Concerns about such costs can be seen at the level of individuals, as in the NOMBA (Not Out of My Bank Account!) phenomenon investigated by Swim et al. (2021). However, such fears are also seen at the corporate and governmental level, with politicians and business leaders sometimes openly questioning whether it makes economic sense to tackle climate change. Climate mitigation and energy transition will be expensive and there are important social justice issues concerning the division of these costs, but the cost of not tackling the climate crisis is many times greater.

Ultimately, humanity's response to the comet in the film fails not because of some inherent species-wide flaw in the ability to understand and think about the problem, but because of the greed of entrenched power and commercial interests: a billionaire subverts the attempt to divert the comet so that he can exploit its presumed mineral wealth. While not a perfect analogy, this parallels the way in which powerful companies have sought to delay climate action so as to continue exploiting fossil fuels for profit. In the film, as in life, GDP is prioritised over preservation of life. The techno-solution of seeking to mine the comet rather than diverting it from its path echoes the arguments that suggest that geo-engineering of our climate can be a substitute for reducing emissions.

Indeed, important aspects of the apparent failure of people to think rationally about climate change are not accidents of human psychology, but rather deliberate manipulations of our psychology by vested interests. Bergquist et al. (2022) report that only 62% of the US population understands that human activities are the primary cause of climate change. This science-public disconnect reflects, in part, the poor communication of scientific topics by mainstream media (also alluded to in the film). However, the problem is not simply poor science communication – it is a deluge of misinformation that causes people to not believe the problem.

Thanks to leaks, freedom of information requests and lawsuits, it is now known that fossil fuel companies deliberately spread climate disinformation for decades. This is not to say that cognitive flaws have not been an important contributor to the inadequate response to the threat of climate change. But these psychological aspects have been exploited and weaponised by corporate interests focused on short-term profit and politicians focused on the short-term electoral cycle. In the film, as in life, humanity has been put in a perilous position because those with power have delayed, denied, and misled.

Approaching comets and the climate crisis are both examples of collective action problems – there is little that individuals can do to meaningfully mitigate against these threats. A critical challenge for psychology is to investigate how to promote the collective action that will be required to bring about a rapid reduction in emissions and a transition to renewable energy. How do we help people to develop a sense of collective efficacy – a belief in their own power to take part in collective action and a belief in the power of collectives to bring about systemic change? Responding to climate change is a problem that requires contributions from a broad range of disciplines, not only in the physical climate sciences, but also in social sciences and the humanities. We need to 'look up' to recognise the impending threat and its real causes, but we also need to look around – outside of our specialist disciplines and across national boundaries – to see how we can harness the power of the collective to mount a truly global response to a global crisis.

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Extract Three:

Adapted from: Lewandowsky, S. (2021). Climate change disinformation and how to combat it. *Annual review of public health*, *42*(1), 1-21.

Notwithstanding the pervasive scientific agreement and measurable adverse consequences of climate change, the public in many countries remain partially unconvinced that climate change presents a societal risk and is caused by fossil-fuel combustion. Climate change presents multiple challenges to human cognition.

One issue is that people systematically underappreciate the implications of carbon emissions. The extent of global warming is determined by the concentration of CO2 in the atmosphere. Reducing emissions will not immediately reduce concentration; it will merely slow the further accumulation of CO2 in the atmosphere. It is only when emissions cease completely, or nearly so, that carbon sinks will begin to draw CO2 from the atmosphere. This is difficult for most people to appreciate, even highly skilled individuals. The mistaken belief that reducing emissions will immediately reduce warming leads people to believe, falsely, that there is still plenty of time, that we can wait and see how bad climate change will be before acting because the climate will respond quickly to policies designed to cut emissions.

Another problem is disinformation. The rhetorical strategies employed by climate disinformation include: undermining and questioning the scientific consensus; highlighting scientific uncertainty and demanding certainty as a condition for climate action; attacking individual scientists to undermine their credibility; undermining institutions generally; and projecting pseudoscientific alternatives through a network of blogs. Disinformation about climate change is disseminated in an organised manner through well-funded networks. For example, analysis of Internal Revenue Service data in the US revealed that the overall annual income of think tanks involved in climate disinformation reached $900 million between 2003 and 2010. At the political level, studies estimate that between 2000 and 2016, more than $2 billion was spent on lobbying Congress concerning climate change legislation, money that was expended primarily on legislators who had an anti-environmental track record.

The media also plays a role. A long-standing journalistic norm is the assumption that there are always two sides to an issue. Accordingly, mainstream media and TV gave contrarian voices equal coverage with climate scientists for a long time. Although the situation has gradually improved, with the media no longer presenting pervasive false balance, the few contrarian scientists continue to be given disproportionate representation. False-balance coverage constitutes one of the most insidious, albeit sometimes inadvertent, forms of climate misinformation. When people see two sides arguing a complicated scientific issue, they come away with the impression of an ongoing, equally split scientific debate. This reduces the public's understanding of the strength of scientific evidence and reduces the public's perception of the scientific consensus.

Calls to educate the public are traditionally made when public attitudes and beliefs diverge from scientific evidence on the assumption that additional information will redress the situation. This deficit model of science communication has been the target of trenchant critique by researchers who highlight the role of people's worldviews and ideologies in the processing of scientific information. On this cultural-cognition view, people are not lacking information but processing it differently. Hence, liberals who endorse climate science are not in possession of more knowledge than conservatives who reject it, but both groups weight information (i.e., the relative risks of unmitigated climate change versus the risks to economic activity from emission cuts) differently. Overall, however, evidence suggests that providing information can be helpful, especially when messages are designed to align with an individual's cultural frame of reference. There are also encouraging signs that education can have long-term effects.

Ultimately, attitudes are notoriously difficult to change, especially if they are central to a person's identity. Researchers have therefore increasingly focused on the communication of specific policies rather than climate change in general. These endeavours have focused mainly on framing climate policies in different ways, for example by underscoring the health benefits that result from emission cuts rather than focusing on environmental benefits. A recent study found framing carbon emissions as "pollution" rather than referring to global warming increased policy support. Communication by scientists and their allies by itself will not move climate mitigation forward. But socioeconomic and political systems are not impervious to change. A Swedish school girl taught the world that lesson in 2019.